

Analysis of the Adverse Health and Economic Impacts of US Storms

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5/10/2017

1: Synopsis

The goal of the assignment is to explore the NOAA Storm Database and explore the effects of severe weather events on both population and economy. The database covers the time period between 1950 and November 2011.

The following analysis investigates which types of severe weather events are most harmful on:

1. Health (injuries and fatalities)
2. Property and crops (economic consequences)

Information on the Data: Documentation

2: Data Processing

2.1: Data Loading

Download the raw data file and extract the data into a dataframe. Then convert to a data.table

```
library("data.table")
library("ggplot2")

fileUrl <- "https://d396qusza40orc.cloudfront.net/repdata%2Fdata%2FStormData.csv.bz2"
destfile <- "StormData.csv.bz2"
download.file(fileUrl, destfile)
stormDF <- read.csv(destfile)

# Converting data.frame to data.table
stormDT <- as.data.table(stormDF)
```

2.2: Examining Column Names

```
colnames(stormDT)

## [1] "STATE_" "BGN_DATE" "BGN_TIME" "TIME_ZONE" "COUNTY"
## [6] "COUNTYNAME" "STATE" "EVTYPE" "BGN_RANGE" "BGN_AZI"
## [11] "BGN_LOCATI" "END_DATE" "END_TIME" "COUNTY_END" "COUNTYENDN"
## [16] "END_RANGE" "END_AZI" "END_LOCATI" "LENGTH" "WIDTH"
## [21] "F" "MAG" "FATALITIES" "INJURIES" "PROPDMG"
## [26] "PROPDMGEXP" "CROPDMG" "CROPDMGEXP" "WFO" "STATEOFFIC"
## [31] "ZONENAMES" "LATITUDE" "LONGITUDE" "LATITUDE_E" "LONGITUDE_"
## [36] "REMARKS" "REFNUM"
```

2.3: Data Subsetting

Subset the dataset on the parameters of interest. Basically, we remove the columns we don't need for clarity.

```

# Finding columns to remove
cols2Remove <- colnames(stormDT[, !c("EVTYPE"
  , "FATALITIES"
  , "INJURIES"
  , "PROPDMG"
  , "PROPDMGEXP"
  , "CROPDMG"
  , "CROPDMGEXP")]))

# Removing columns
stormDT[, c(cols2Remove) := NULL]

# Only use data where fatalities or injuries occurred.
stormDT <- stormDT[(EVTYPE != "?" &
  (INJURIES > 0 | FATALITIES > 0 | PROPDMG > 0 | CROPDMG > 0)), c("EVTYPE"
  , "FATALITIES"
  , "INJURIES"
  , "PROPDMG"
  , "PROPDMGEXP"
  , "CROPDMG"
  , "CROPDMGEXP") ]

```

2.4: Converting Exponent Columns into Actual Exponents instead of (-,+, H, K, etc)

Making the PROPDMGEXP and CROPDMGEXP columns cleaner so they can be used to calculate property and crop cost.

```

# Change all damage exponents to uppercase.
cols <- c("PROPDMGEXP", "CROPDMGEXP")
stormDT[, (cols) := c(lapply(.SD, toupper)), .SDcols = cols]

# Map property damage alphanumeric exponents to numeric values.
propDmgKey <- c("\\" = 10^0,
  "-" = 10^0,
  "+" = 10^0,
  "0" = 10^0,
  "1" = 10^1,
  "2" = 10^2,
  "3" = 10^3,
  "4" = 10^4,
  "5" = 10^5,
  "6" = 10^6,
  "7" = 10^7,
  "8" = 10^8,
  "9" = 10^9,
  "H" = 10^2,
  "K" = 10^3,
  "M" = 10^6,
  "B" = 10^9)

# Map crop damage alphanumeric exponents to numeric values
cropDmgKey <- c("\\" = 10^0,
  "?" = 10^0,
  "0" = 10^0,

```

```

      "K" = 10^3,
      "M" = 10^6,
      "B" = 10^9)

stormDT[, PROPDMGEXP := propDmgKey[as.character(stormDT[,PROPDMGEXP])]]
stormDT[is.na(PROPDMGEXP), PROPDMGEXP := 10^0 ]

stormDT[, CROPDMGEXP := cropDmgKey[as.character(stormDT[,CROPDMGEXP])]] ]
stormDT[is.na(CROPDMGEXP), CROPDMGEXP := 10^0 ]

```

2.5: Making Economic Cost Columns

```

stormDT <- stormDT[, .(EVTYPE, FATALITIES, INJURIES, PROPDMG, PROPDMGEXP, propCost = PROPDMG * PROPDMGEXP,

```

2.6: Calculating Total Property and Crop Cost

```

totalCostDT <- stormDT[, .(propCost = sum(propCost), cropCost = sum(cropCost), Total_Cost = sum(propCost + cropCost))]
totalCostDT <- totalCostDT[order(-Total_Cost), ]
totalCostDT <- totalCostDT[1:10, ]

head(totalCostDT, 5)

```

##	EVTYPE	propCost	cropCost	Total_Cost
##	<char>	<num>	<num>	<num>
## 1:	FLOOD	144657709807	5661968450	150319678257
## 2:	HURRICANE/TYPHOON	69305840000	2607872800	71913712800
## 3:	TORNADO	56947380676	414953270	57362333946
## 4:	STORM SURGE	43323536000	5000	43323541000
## 5:	HAIL	15735267513	3025954473	18761221986

2.7: Calculating Total Fatalities and Injuries

```

totalInjuriesDT <- stormDT[, .(FATALITIES = sum(FATALITIES), INJURIES = sum(INJURIES), totals = sum(FATALITIES + INJURIES))]
totalInjuriesDT <- totalInjuriesDT[order(-FATALITIES), ]
totalInjuriesDT <- totalInjuriesDT[1:10, ]

head(totalInjuriesDT, 5)

```

##	EVTYPE	FATALITIES	INJURIES	totals
##	<char>	<num>	<num>	<num>
## 1:	TORNADO	5633	91346	96979
## 2:	EXCESSIVE HEAT	1903	6525	8428
## 3:	FLASH FLOOD	978	1777	2755
## 4:	HEAT	937	2100	3037
## 5:	LIGHTNING	816	5230	6046

3: Results

3.1: Events that are Most Harmful to Population Health

Melting data.table so that it is easier to put in bar graph format

```
bad_stuff <- melt(totalInjuriesDT, id.vars="EVTTYPE", variable.name = "bad_thing")
head(bad_stuff, 5)
```

```
##           EVTTYPE bad_thing value
##           <char>      <fctr> <num>
## 1:    TORNADO FATALITIES  5633
## 2: EXCESSIVE HEAT FATALITIES 1903
## 3:   FLASH FLOOD FATALITIES  978
## 4:         HEAT FATALITIES  937
## 5:   LIGHTNING FATALITIES  816
```

```
# Create chart
```

```
healthChart <- ggplot(bad_stuff, aes(x=reorder(EVTTYPE, -value), y=value))
```

```
# Plot data as bar chart
```

```
healthChart = healthChart + geom_bar(stat="identity", aes(fill=bad_thing), position="dodge")
```

```
# Format y-axis scale and set y-axis label
```

```
healthChart = healthChart + ylab("Frequency Count")
```

```
# Set x-axis label
```

```
healthChart = healthChart + xlab("Event Type")
```

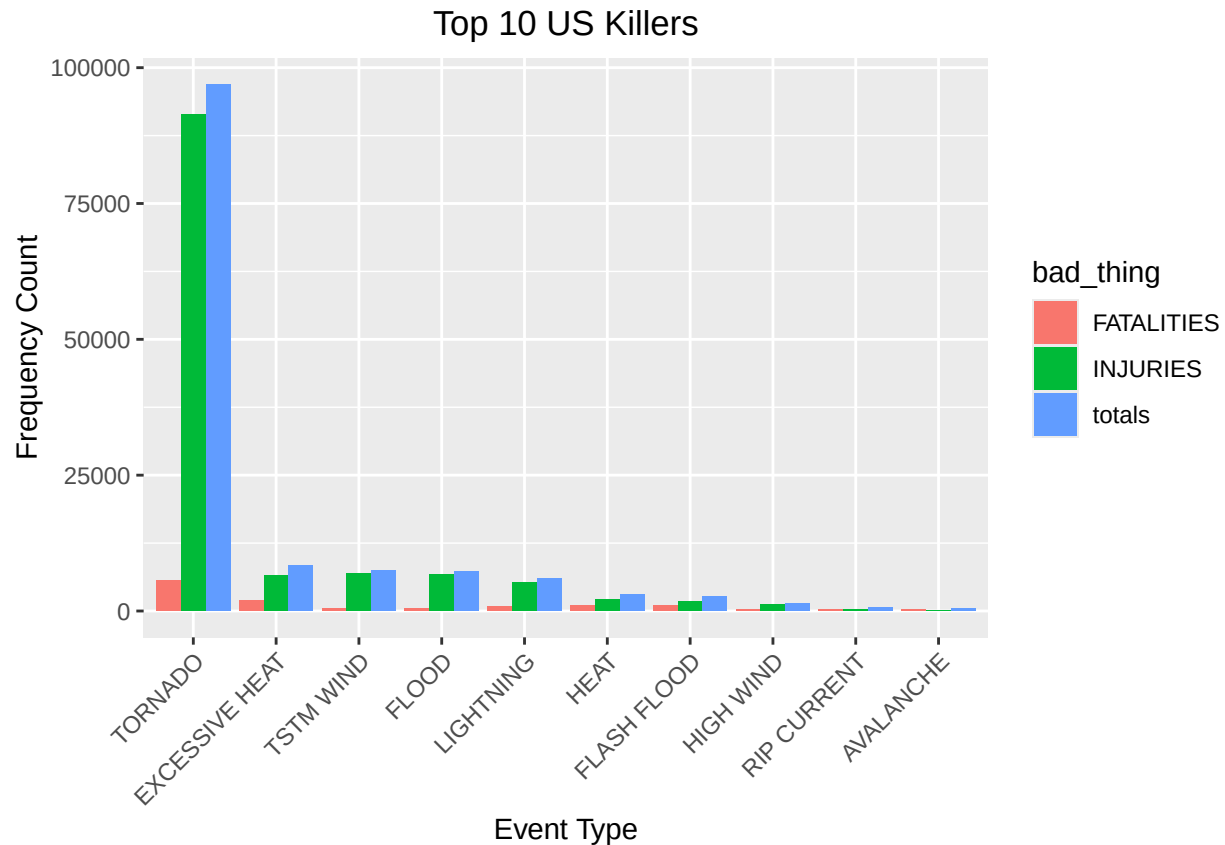
```
# Rotate x-axis tick labels
```

```
healthChart = healthChart + theme(axis.text.x = element_text(angle=45, hjust=1))
```

```
# Set chart title and center it
```

```
healthChart = healthChart + ggtitle("Top 10 US Killers") + theme(plot.title = element_text(hjust = 0.5))
```

```
healthChart
```



3.2: Events that have the Greatest Economic Consequences

Melting data.table so that it is easier to put in bar graph format

```
econ_consequences <- melt(totalCostDT, id.vars="EVTYPE", variable.name = "Damage_Type")
head(econ_consequences, 5)
```

```
##           EVTYPE Damage_Type      value
##           <char>      <fctr>      <num>
## 1:           FLOOD      propCost 144657709807
## 2: HURRICANE/TYPHOON      propCost  69305840000
## 3:           TORNADO      propCost  56947380676
## 4:    STORM SURGE      propCost  43323536000
## 5:           HAIL      propCost  15735267513
```

Create chart

```
econChart <- ggplot(econ_consequences, aes(x=reorder(EVTYPE, -value), y=value))
```

Plot data as bar chart

```
econChart = econChart + geom_bar(stat="identity", aes(fill=Damage_Type), position="dodge")
```

Format y-axis scale and set y-axis label

```
econChart = econChart + ylab("Cost (dollars)")
```

Set x-axis label

```
econChart = econChart + xlab("Event Type")
```

```

# Rotate x-axis tick labels
econChart = econChart + theme(axis.text.x = element_text(angle=45, hjust=1))

# Set chart title and center it
econChart = econChart + ggtitle("Top 10 US Storm Events causing Economic Consequences") + theme(plot.title = element_text(hjust = 0.5))

econChart

```

